Exploring Preterm Births with K-Means and Other ML Techniques

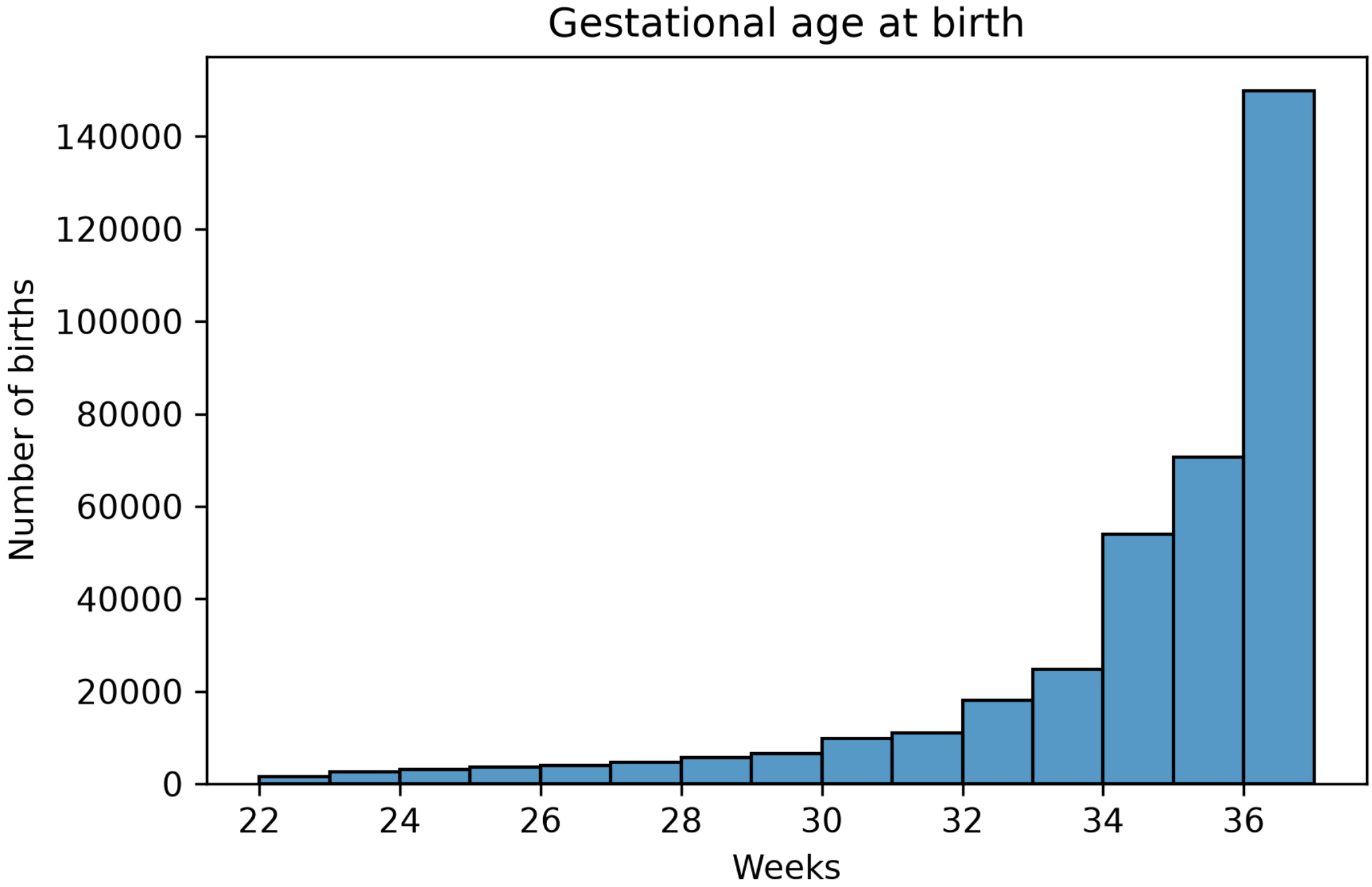
AI in Healthcare

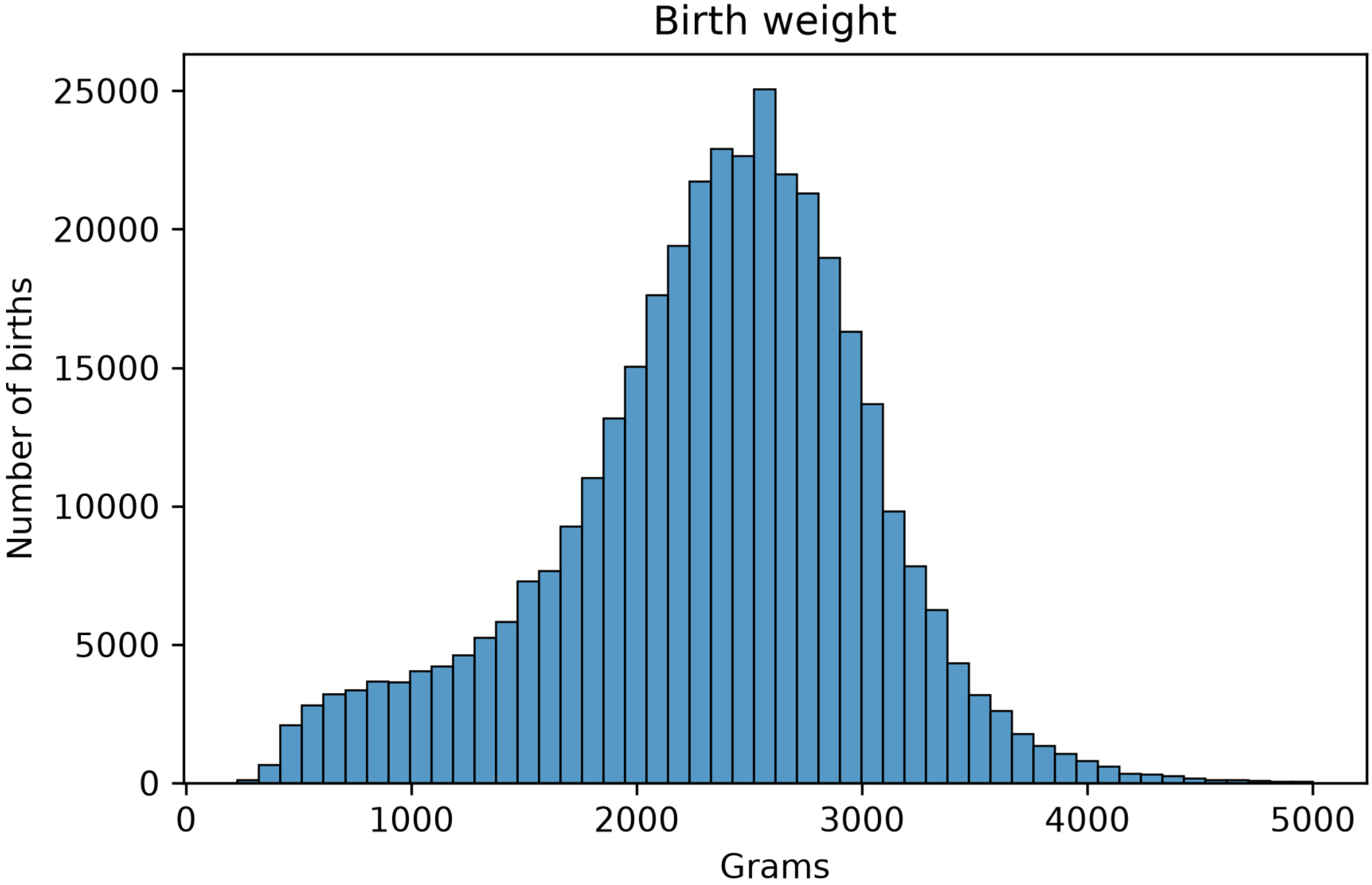
Self Learning Tutorial

10/2026

Motivation

- A preterm or premature birth is defined as a birth of a baby before 37 weeks
- In the US, approximately 10% - 11% of births are preterm
- Premature birth can have a wide range of outcomes and impacts on those children and their families
- CDC's linked infant birth/death records has data on preterm births.
- Goal: can we look at the data to group different types of preterm births and measure what, if any, different outcomes they face.
- Use ML methods in this case


Gestational age at birth
Number of births
140000
120000
100000
80000
60000
40000
20000
0
22
24
26
28
30
32
34
36
Weeks


Birth weight
Number of births
25000
20000
15000
10000
5000
0
0
1000
2000
3000
4000
5000
Grams

CDC Linked Infant Birth and Death Records

- Our basis for analysis is the CDC 2023 cohort of infant birth and death records (CDC tracks all deaths of infant before the age of 1).
- The data can be found from the US CDC National Center for Health Statistics: https://ftp.cdc.gov/pub/Health_Statistics/NCHS/Datasets/DVS/period-cohort-linked/2024PE2023CO.zip
- For US residents, we saw that there were > 373K preterm births in 2023 (out of > 3.6M births total), and we ultimately looked at > 367K of those records
- Roughly 9.5K of those records that we looked at ended in a death before the age of 1

Reading in the Data

- Each row in the data represents one birth, but contains no commas or column headers
- Instead the field values live in fixed positions in the rows
- We use the CDC user guide for this dataset to develop this schema and store it in a dictionary called FIELDS.
- User guide: https://ftp.cdc.gov/pub/Health_Statistics/NCHS/Dataset_Documentation/DVS/period-cohort-linked/24PE23CO_linkedUG.pdf

```
FIELDS = {
    "restatus": (104, 104), # residency status, 4 = foreign resident (exclude foreign residents)
    "oegest_r3": (503, 503), # preterm flag, 1 = under 37 weeks (preterm)
    "mager": (75, 76), # mother's age in years
```

```
colspecs = [(start - 1, end) for start, end in FIELDS.values()]
chunks = pd.read_fwf(DENOM_PATH, colspecs=colspecs, names=list(FIELDS),
                     dtype=str, encoding="latin-1", chunksize=500000)
```

75-76	2	P,G	MAGER	**Mother's Age Recode 41**	12 10 – 12 years, 13 13 years, 14 14 years, 15 15 years, 16 16 years, 17 17 years, 18 18 years, 19 19 years, 20 20 years, 21 21 years, 22 22 years, 23 23 years, 24 24 years, 25 25 years, 26 26 years, 27 27 years, 28 28 years, 29 29 years, 30 30 years, 31 31 years, 32 32 years, 33 33 years, 34 34 years, 35 35 years, 36 36 years, 37 37 years, 38 38 years, 39 39 years, 40 40 years, 41 41 years, 42 42 years, 43 43 years, 44 44 years, 45 45 years, 46 46 years, 47 47 years, 48 48 years, 49 49 years, 50 50 years and over

Data Cleansing

- Once we load the data we make some changes, e.g., convert numbers stored as text into numerical objects. Unknown values are given special values in the user guide and we apply those values to filter out rows where necessary.
- Example: Yes and No fields are assigned to 1 and 0, and unknown values are assigned NaN.
- Infant deaths are tagged as 1 if the birth record can be linked to a death record. Otherwise it is tagged as 0.
- All records where the birth weight was guessed are dropped

Linking Birth and Death Records

- For a baby born in 2023, its death can occur in 2023 or 2024, so we need the two death certificate files in 2024PE2023CO
- Death records share two values with birth records, a sequence number and year of death. We use both of these to identify a link since sequence numbers are reset each year (and thus do not link a birth to a death on their own)
- We merge the data with the age at death and cause of death whenever applicable to the corresponding birth

```
df["days"] = df["aged"].fillna(365)
df["event"] = df["infant_died"]

# Check: matched deaths should equal (or nearly equal) the infant_died count
print("Number of deaths flagged in births file: ", df['infant_died'].sum())
print("Number of deaths with age at death: ", df['aged'].notna().sum())
```

```
Number of deaths flagged in births file:  11993
Number of deaths with age at death:  11993
```

Data Validation

```
print("Infant deaths: ", df['infant_died'].sum())
df["gest_group"] = pd.cut(df["oegest"], bins=[0, 27, 31, 33, 36],
                          labels=["<28 wks", "28-31", "32-33", "34-36"])
print((df.groupby("gest_group", observed=True)["infant_died"].mean() * 1000).round(1))
```

```
Infant deaths:  11993
gest_group
<28 wks    338.0
28-31       42.7
32-33       20.2
34-36        8.1
Name: infant_died, dtype: float64
```

	Gestation									
Birthweight	Total	<28 Weeks	28-31 Weeks	32-33 Weeks	34-36 Weeks	37-39 Weeks	40 Weeks	41 Weeks	42 Weeks or more	Not Stated
All Races										
Live births	3,596,017	23,096	33,153	43,006	274,647	2,421,849	624,543	163,782	9,299	2,642
Infant deaths	19,743	8,246	1,428	877	2,235	5,641	859	224	32	201
Infant Mortality Rate	5.49	357.03	43.07	20.39	8.14	2.33	1.38	1.37	3.44	76.08

Filtering the Data

- In our analysis we started with > 3.6M births.
- Our focus on preterm births to US residents resulted in us keeping only 373902 rows in the data that correspond to this filter.
- From there, we remove preterm births before 22 weeks and remove births where the birthweight was > ~11 lbs, which left 370003 rows in the data. This also removed >2500 deaths associated with preterm babies <22 weeks old.
- Finally, we removed rows that have missing feature data. All in we looked at 367081 birth cases.

Choice of Cluster Features

Rules for cluster choices:

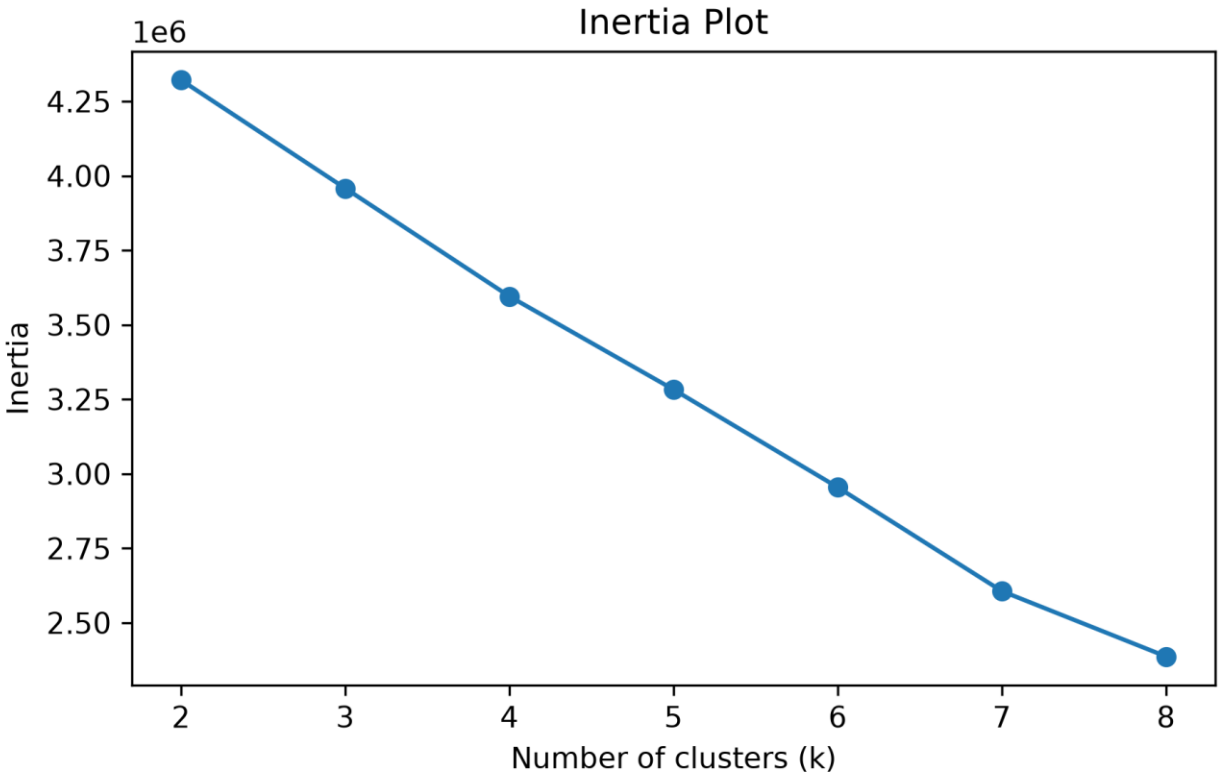
1. Features must have been known at or before birth
2. Feature had to be a known risk factor for a preterm birth

These features chosen were as follows: gestational age, birth weight, whether delivery was for more than one child, e.g., twins, pre-pregnancy hypertension, gestational hypertension, eclampsia, pre-pregnancy diabetes, gestational diabetes, infection (Chorioamnionitis), previous preterm birth, fertility treatments, maternal age, and smoking

Choice of K

- For practical purposes we look at values of K = 2, 3, …, 8 and then make a decision on a “good” value for K.
- After we test run K-Means clustering for those different values for K, we look at two measures of quality: Inertia and Silhouette scores.

Choose K = 7


Inertia Plot
1e6
4.25
4.00
3.75
3.50
3.25
3.00
2.75
2.50
Inertia
2
3
4
5
6
7
8
Number of clusters (k)


Silhouette score
0.5
0.4
0.3
0.2
0.1
0.0
Silhouette
2
3
4
5
6
7
8
Number of clusters (k)

7 Chosen Clusters

- Very preterm, unexplained
- Infection associated
- Pre-existing diabetes
- Fertility treatment
- Eclampsia
- Recurrent preterm delivery
- Late preterm, unexplained.

```
# Here we fit the final k-means model with the chosen number of clusters based on the graph (use 7 in this case)

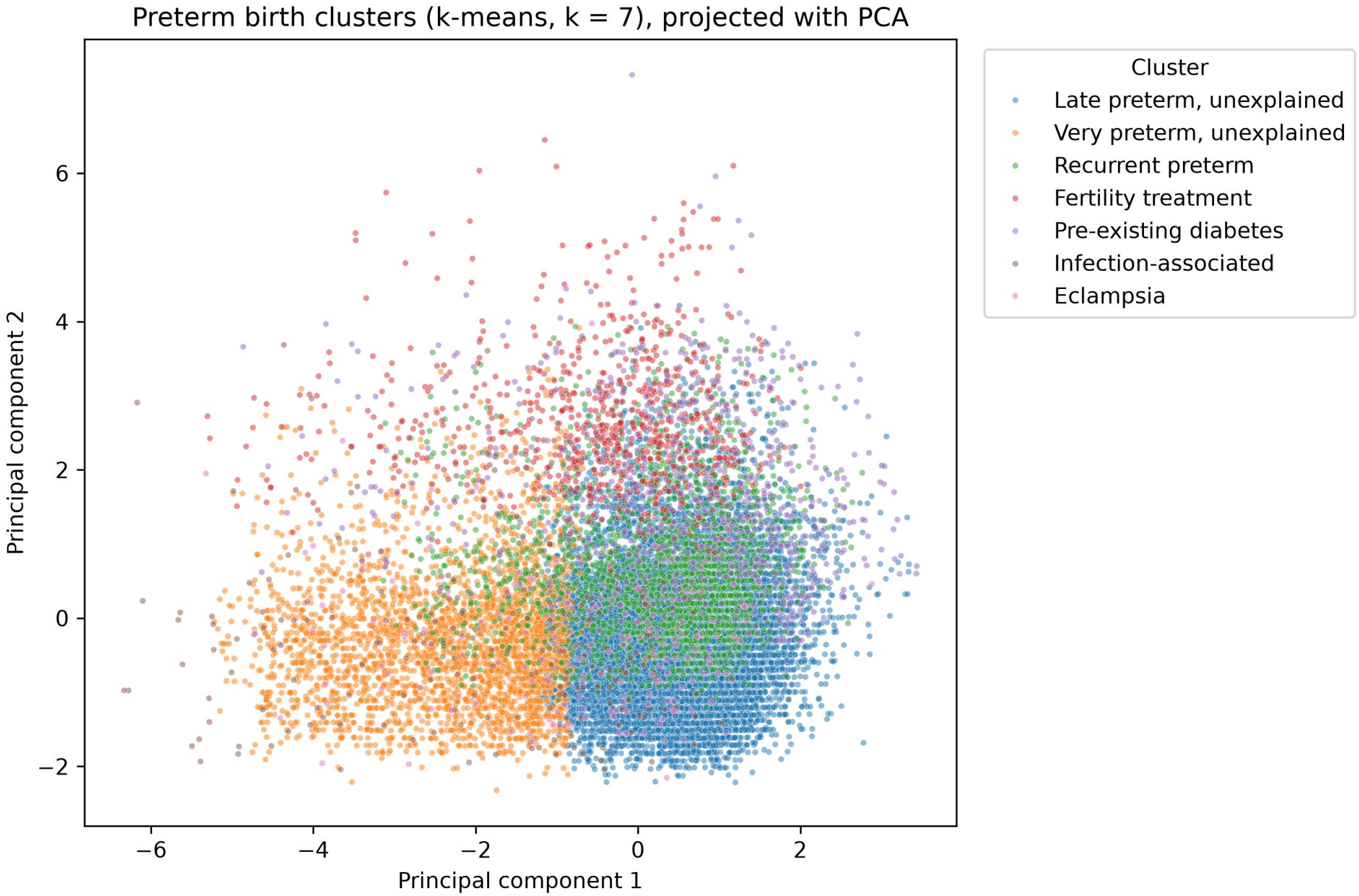
K = 7
kmeans = KMeans(n_clusters=K, n_init=10, random_state=42).fit(X)
model_df["cluster"] = kmeans.labels_
profile = model_df.groupby("cluster")[FEATURES + OUTCOMES].mean()
profile.insert(0, "n_births", model_df["cluster"].value_counts().sort_index())

profile["deaths_per_1000"] = profile["infant_died"] * 1000
profile.round(2).T
```

cluster	0	1	2	3	4	5	6
n_births	53530.00	4443.00	35784.00	12256.00	14772.00	242929.00	3367.00
oegest	28.89	31.61	34.35	33.97	33.71	35.12	33.58
brthwgt	1244.15	1964.07	2405.69	2651.45	2229.01	2564.00	2175.91
multiple_birth	0.28	0.14	0.14	0.10	0.43	0.17	0.13
mager	29.26	29.49	31.09	31.73	35.32	29.68	29.53
rf_pdiab	0.00	0.04	0.00	1.00	0.00	0.00	0.06
rf_gdiab	0.07	0.11	0.14	0.00	0.17	0.12	0.12
rf_phype	0.08	0.05	0.10	0.28	0.08	0.05	0.09
rf_ghype	0.20	0.14	0.18	0.29	0.27	0.20	0.16
rf_ehype	0.00	0.00	0.00	0.00	0.00	0.00	1.00
rf_ppb	0.04	0.10	1.00	0.18	0.07	0.00	0.08
rf_inft	0.00	0.05	0.00	0.03	1.00	0.00	0.04
ld_chor	0.00	1.00	0.00	0.00	0.00	0.00	0.00
cig_rec	0.05	0.04	0.08	0.05	0.00	0.04	0.05
infant_died	0.11	0.08	0.01	0.02	0.02	0.01	0.02
ab_nicu	0.90	0.70	0.54	0.64	0.61	0.42	0.63
ab_aven6	0.32	0.26	0.15	0.20	0.19	0.08	0.16
ab_surf	0.15	0.11	0.04	0.06	0.07	0.01	0.06
deaths_per_1000	108.46	79.90	11.65	24.23	21.39	9.18	17.52

Use PCA to Visualize Clusters

- Principal component analysis (PCA) allows us to linearly project onto lower dimensional space to see the directions along which the data varies the most.
- In our analysis, along the first axis variation stems from when the baby arrived (early/late preterm) together with its birth weight
- Along the second axis, the biggest drivers are maternal age and the mother's health, e.g., infertility treatment, pre-pregnancy hypertension, etc.
- Some features like Eclampsia or gestational hypertension don't register in the PCA chart


Preterm birth clusters (k-means, k = 7), projected with PCA
Principal component 2
Principal component 1
6
4
2
0
−2
−6
−4
−2
0
2
Cluster
Late preterm, unexplained
Very preterm, unexplained
Recurrent preterm
Fertility treatment
Pre-existing diabetes
Infection-associated
Eclampsia

Gaussian Mixture Model

- K-Means will give us the clusters we ask for (as long as we have at least K datapoints).
- How do we know if we captured the right clusters?
- We test the same data using a Gaussian Mixture Model to give us 7 clusters.
- GMM models each group as a normal distribution and gives a confidence level for each assignment
- Compare the results using an Adjusted Rand Index

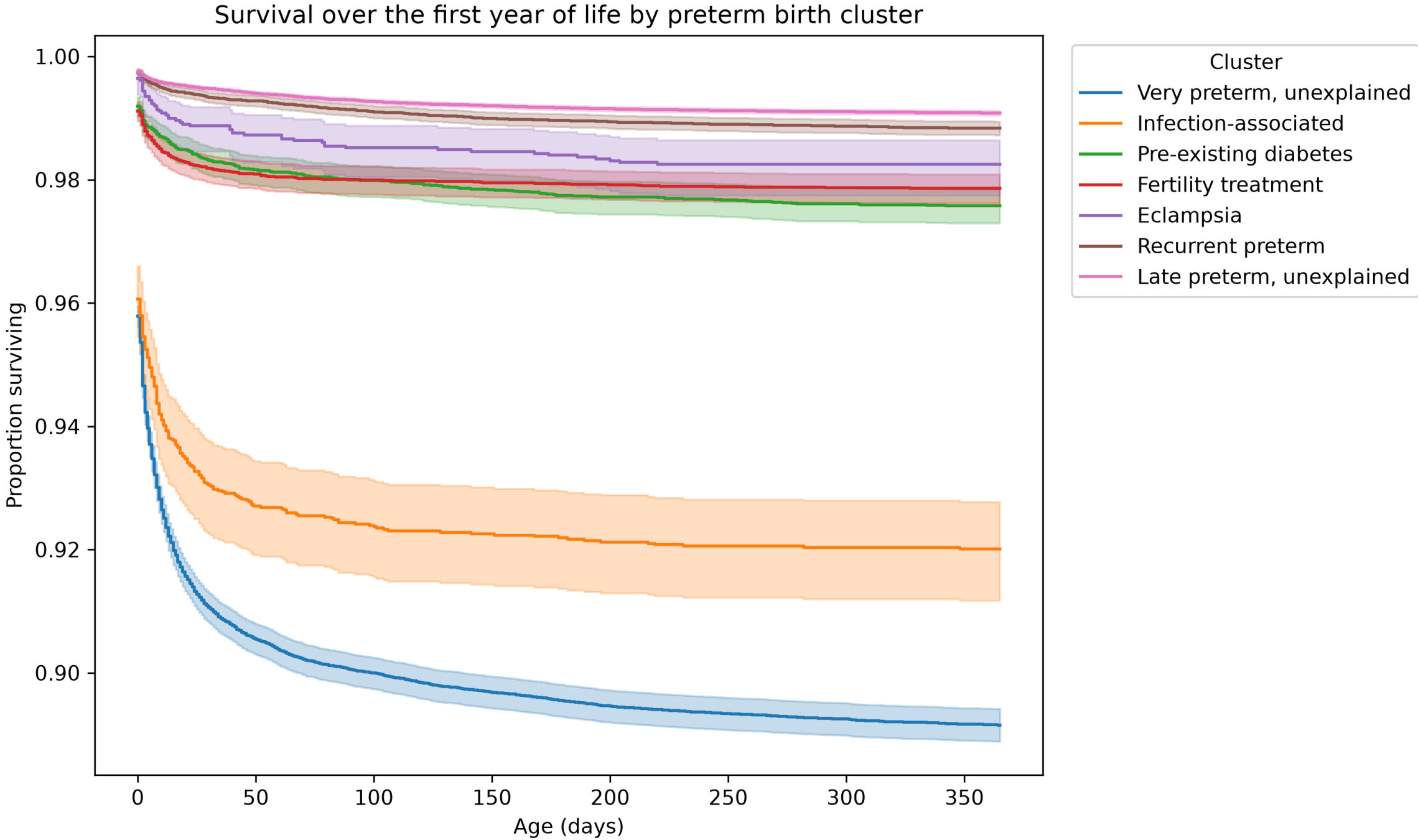
GMM Results + ARI

```
Median assignment confidence:  1.0
Adjusted Rand Index (k-means vs GMM):  0.18
```

gmm_cluster	0	1	2	3	4	5	6
cluster_name							
Eclampsia	100.0	0.0	0.0	0.0	0.0	0.0	0.0
Fertility treatment	0.0	0.0	0.0	33.4	49.4	17.2	0.0
Infection-associated	0.0	0.0	0.0	0.0	100.0	0.0	0.0
Late preterm, unexplained	0.0	13.6	53.2	14.0	0.0	11.9	7.3
Pre-existing diabetes	0.0	28.3	0.0	0.0	3.3	0.0	68.4
Recurrent preterm	0.0	12.4	0.0	10.0	0.0	13.8	63.7
Very preterm, unexplained	0.0	15.0	43.6	22.9	0.0	6.5	11.9

Survivability Analysis of the K-Means Cohorts

- Since we could not generate wider agreement between our K-Means and GMM models, we stick with the K-Means clusters.
- Then we want to understand how babies in each cluster survive during the first year
- We use Kaplan-Meier curves to show this


Survival over the first year of life by preterm birth cluster
Proportion surviving
1.00
0.98
0.96
0.94
0.92
0.90
0
50
100
150
200
250
300
350
Age (days)
Cluster
Very preterm, unexplained
Infection-associated
Pre-existing diabetes
Fertility treatment
Eclampsia
Recurrent preterm
Late preterm, unexplained

Causes of Death in the Cluster

cluster_name cause	Eclampsia	Fertility treatment	Infection-associated	Late preterm, unexplained	Pre-existing diabetes	Recurrent preterm	Very preterm, unexplained
Birth defects	11.9	14.2	8.2	43.5	23.6	31.7	13.7
Injury	1.7	0.0	0.0	3.1	1.0	3.6	0.6
Newborn infections	6.8	6.3	14.1	1.5	3.4	2.6	9.1
Other	10.2	8.2	8.5	12.9	12.8	13.9	9.5
Other newborn conditions	37.3	43.0	31.3	13.1	34.3	18.5	39.6
Pregnancy / delivery complications	5.1	9.5	19.2	3.7	7.1	6.7	9.6
Short gestation / low birth weight	13.6	17.1	15.5	0.5	8.8	1.0	15.3
Sudden unexpected infant death	13.6	1.6	3.4	21.6	9.1	22.1	2.5

Design Limitations

- We made several assumptions about the data, e.g., ignoring preterm babies < 22 weeks old and babies > 11lbs at birth, which removed > 2500 death links
- Clinical data often overlaps but this often leads to lower Silhouette scores, which we saw when we did the K-Means cluster analysis
- We could have also chosen different top values for K, e.g., > 8.
- The nature of our feature sets (many Y/N, e.g., 0 or 1 features) tended to form their own clusters, which made the GMM less robust as a method.

Future Work

- Work with MIMIC-III data, which also has information on preterm births
- Use other clustering algorithms
- Use social factors in our analysis, e.g., education, race, insurance type, etc., which can have an impact on preterm births

Conclusions

- Can use unsupervised ML methods to cluster the data
- Clusters are highly sensitive to the methods used, features, and choices/assumptions made by us (the practitioners)
- Overall, gestational age, e.g., how early or late the baby was born, was the most strongly associated with survivability.

Bibliography & Hardware/Software Versions

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